A2 1: INFORMATION SYSTEMS · RELATIONAL DATABASE DESIGN

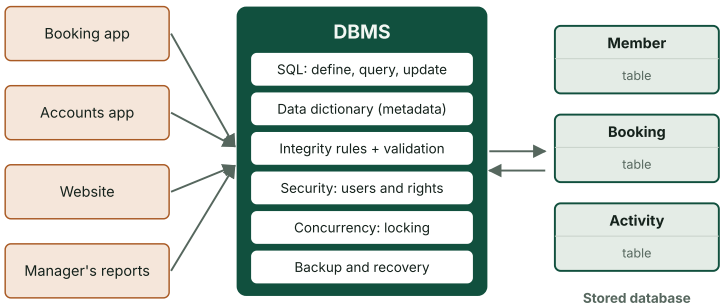
Data management: flat files, relational databases and the DBMS

IS01 WORKSHEET

Name	Class	Date	Score

I can: Explain why flat files lead to data redundancy and data inconsistency · Describe at least five functions of a DBMS · Explain what data independence means and why it matters to an organisation

All access to the data goes through the DBMS



Data independence: storage can change without rewriting the programs, because they use the schema

Use this diagram to help you: Application programs reach the data only through the DBMS, which provides SQL, the data dictionary, integrity, security, concurrency control and backup; this gives data independence

A Key terms

Write the letter of the matching definition · 6 marks

- ☐ **Flat file**
- ☐ **Data redundancy**
- ☐ **Data inconsistency**
- ☐ **Database management system (DBMS)**
- ☐ **Data independence**
- ☐ **Relational database**

A A database held as a single table or file with no relationships to other tables.

B A situation where copies of the same data item hold different values, so it is unclear which is correct.

C Software that allows a database to be defined, created, queried, updated and controlled, including security, integrity and concurrent access.

D The unnecessary storage of the same data item more than once.

E A database in which data is stored in tables (relations) linked to each other through primary and foreign keys.

F The ability to change the way data is stored or structured without having to change the application programs that use it.

B Fill the gaps

Use the word bank · 5 marks

tables | DBMS | inconsistency | independence | redundancy

In a flat file the same data is often stored many times, which is called data _____. If one copy is changed and another is not, the result is data _____. A relational database splits data into linked _____. The software that controls access to the database is the _____. Changing how data is stored without rewriting programs is known as data _____.

C True or false?

Tick one box · 6 marks

	True	False
A flat file stores data in a single table with no links to other tables.	☐	☐
Data redundancy and data inconsistency mean the same thing.	☐	☐
A DBMS keeps a data dictionary describing the structure of the database.	☐	☐
Data independence means programs must be rewritten when a field is added.	☐	☐
A relational database removes all redundancy completely.	☐	☐
Record locking is one way a DBMS manages concurrent access.	☐	☐

D Exam-style questions

16 marks

1 Define the term data redundancy. [1]

2 Explain how data redundancy can lead to data inconsistency. [2]

3 Describe four functions of a database management system. [4]

4 Explain what is meant by data independence and give one benefit to an organisation. [3]

5 Erne Leisure Centre stores all bookings in a single flat file. Discuss the benefits and drawbacks of replacing the flat file with a relational database managed by a DBMS. [6]

E Quick quiz

Circle one answer · 5 marks

1 Which problem occurs when two copies of a customer's address hold different values? [1]

A Data redundancy
B Data inconsistency
C Data independence
D Referential integrity

2 Which is NOT normally a function of a DBMS? [1]

A Enforcing access rights
B Maintaining a data dictionary
C Designing the ER diagram for you
D Backup and recovery

3 Adding an index without changing any application program is an example of... [1]

A logical data independence
B program-data dependence
C physical data independence
D data redundancy

4 In a relational database, tables are linked using... [1]

A foreign keys
B validation rules
C macros
D indexes only

5 When is a flat file most appropriate? [1]

A A national bank's accounts
B A hospital patient system
C A small single-user contact list
D An airline booking system

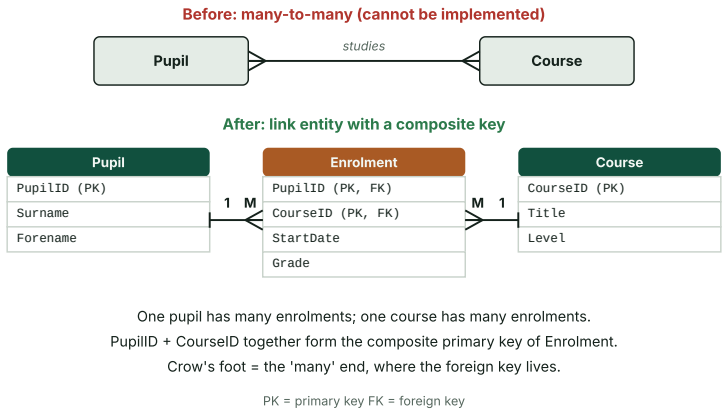
A2 1: INFORMATION SYSTEMS · RELATIONAL DATABASE DESIGN

Entity-relationship modelling

IS02 WORKSHEET

Name	Class	Date	Score

I can: Identify entities, attributes and keys from a scenario · Draw an ER diagram using crow's foot notation with the correct degree of each relationship · Resolve a many-to-many relationship using a link entity with a composite key



Use this diagram to help you: A many-to-many relationship between Pupil and Course resolved by an Enrolment link entity with a composite key, giving two one-to-many relationships

A Key terms

Write the letter of the matching definition · 6 marks

- ☐ **Entity**
- ☐ **Attribute**
- ☐ **Primary key**
- ☐ **Foreign key**
- ☐ **Composite key**
- ☐ **Link entity**

A An attribute, or combination of attributes, that uniquely identifies each record in a table.

B A property or characteristic of an entity; it becomes a field in a table.

C An object, person, place or event about which data is stored; it becomes a table in a relational database.

D An attribute in one table that is the primary key of another table and is used to link the two tables.

E An entity introduced to resolve a many-to-many relationship into two one-to-many relationships.

F A primary key made up of two or more attributes which together uniquely identify a record.

B Fill the gaps

Use the word bank · 5 marks

link | attributes | many-to-many | composite | entity

A thing about which data is stored is called an _____. Its properties are called _____. A key made from two or more attributes is a _____ key. A _____ relationship must be resolved by adding a _____ entity.

C True or false?

Tick one box · 5 marks

	True	False
In a one-to-many relationship the foreign key goes in the table at the 'many' end.	☐	☐

Entity names in an ER diagram should be plural, e.g. Customers.	☐	☐
A link entity changes one M:N relationship into two 1:M relationships.	☐	☐
A foreign key must be unique within its own table.	☐	☐
A candidate key is any attribute that could uniquely identify a record.	☐	☐

D Exam-style questions

14 marks

1 State what is meant by a composite key. [1]

2 Explain the purpose of a foreign key. [2]

3 A library allows each member to borrow many books, and over time each book is borrowed by many members. Draw an ER diagram showing how this relationship would be implemented in a relational database. [4]

4 Write an entity description for the link entity in your answer above, identifying its key(s). [3]

5 Explain why a many-to-many relationship cannot be implemented directly in a relational database. [2]

6 Identify the degree of the relationship between a Translink bus route and its stops, and justify your answer. [2]

E Quick quiz

Circle one answer · 5 marks

1 Which relationship needs a link entity? [1]

A One-to-one
B One-to-many
C Many-to-many
D None of these

2 Enrolment(PupilID, CourseID, Grade) uses PupilID and CourseID together as... [1]

A a secondary key
B a composite key
C a candidate attribute
D an index only

3 Where is the crow's foot drawn in Customer ——< Order? [1]

A At Customer
B At Order
C At both ends
D Nowhere

4 A head of department manages exactly one department, and each department has one head. This is... [1]

A 1:M
B M:N
C M:1
D 1:1

5 Which statement about a foreign key is correct? [1]

A It must be the first field in a table
B It is the primary key of another table
C It cannot contain repeated values
D It is always a composite key

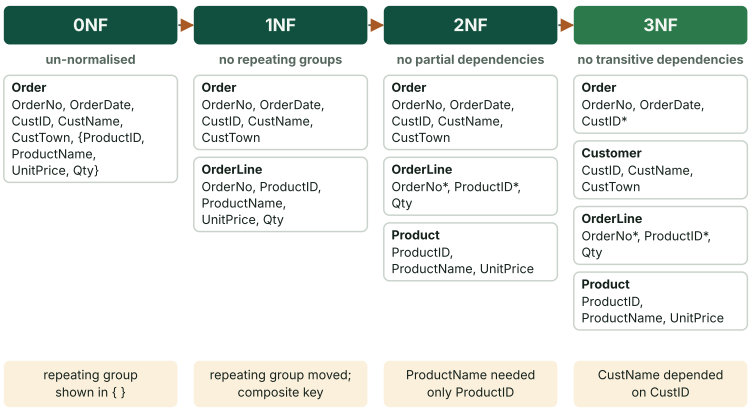
A2 1: INFORMATION SYSTEMS · RELATIONAL DATABASE DESIGN

Normalisation to third normal form

IS03 WORKSHEET

Name	Class	Date	Score

I can: Explain why normalisation is used and the problems (anomalies) it prevents · Take a set of data from 0NF to 3NF, showing each stage · Identify repeating groups, partial dependencies and transitive dependencies



Use this diagram to help you: Fermanagh Fresh order data normalised from 0NF through 1NF and 2NF to 3NF, showing which dependency is removed at each stage

A Key terms

Write the letter of the matching definition · 6 marks

- ☐ **Normalisation**
- ☐ **First normal form (1NF)**
- ☐ **Second normal form (2NF)**
- ☐ **Third normal form (3NF)**
- ☐ **Partial dependency**
- ☐ **Transitive dependency**

A A relation in 2NF with no transitive dependencies, so every non-key attribute depends only on the primary key.

B A relation with no repeating groups in which every attribute holds a single atomic value.

C Where a non-key attribute depends on another non-key attribute rather than directly on the primary key.

D A relation in 1NF with no partial dependencies on a composite key.

E A formal process of organising data into relations to minimise redundancy and prevent update, insertion and deletion anomalies.

F Where a non-key attribute depends on only part of a composite primary key.

B Fill the gaps

Use the word bank · 5 marks

composite · redundancy · transitive · partial · repeating

A relation is in first normal form when it has no _____ groups. To reach second normal form we remove _____ dependencies, which can only exist when the table has a _____ key. Third normal form removes _____ dependencies. Normalisation reduces data _____.

C True or false?

Tick one box · 6 marks

	True	False

A table with a single-attribute primary key that is in 1NF is automatically in 2NF.	☐	☐
A transitive dependency is where a non-key attribute depends on part of the key.	☐	☐
At 1NF the repeating group moves to a new table along with the original key.	☐	☐
Normalisation increases data redundancy.	☐	☐
Derived values such as a line total are usually removed during normalisation because they can be calculated.	☐	☐
A fully normalised database never needs joins.	☐	☐

D Exam-style questions

18 marks

1 State two reasons why data is normalised. [2]

2 Define a partial dependency. [2]

3 Define a transitive dependency and give an example. [2]

4 Student(StudentID, Name, {CourseCode, CourseTitle, TutorID, TutorName, Grade}). Normalise this relation to third normal form, showing each stage. [6]

5 Explain what is meant by an insertion anomaly, using an example. [2]

6 Evaluate whether a database should always be normalised to third normal form. [4]

E Quick quiz

Circle one answer · 5 marks

1 Which form removes repeating groups? [1]

A 1NF
B 2NF
C 3NF
D 0NF

2 OrderLine(OrderNo, ProductID, ProductName, Qty): ProductName depends on ProductID only. This is a... [1]

A transitive dependency
B partial dependency
C repeating group
D full dependency

3 Order(OrderNo, CustID, CustTown): CustTown depends on CustID. Removing this gives... [1]

A 1NF
B 2NF
C 3NF
D 0NF

4 Deleting the only order for a customer also loses the customer's address. This is a... [1]

A update anomaly
B insertion anomaly
C referential anomaly
D deletion anomaly

5 The phrase 'the key, the whole key and nothing but the key' summarises... [1]

A SQL joins
B 3NF
C record locking
D ER diagrams

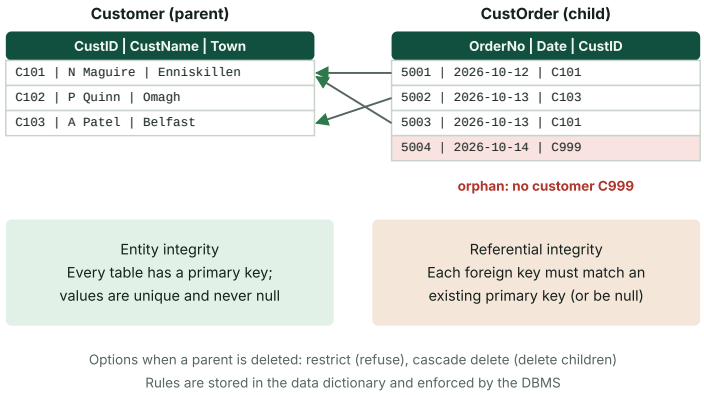
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Data integrity: data dictionary, integrity rules, validation and indexing

IS04 WORKSHEET

Name	Class	Date	Score

I can: Describe what a data dictionary stores and who uses it · Explain entity and referential integrity, including cascade update and delete · Evaluate the use of indexes in a relational database



Use this diagram to help you: Entity integrity requires unique, non-null primary keys; referential integrity requires every foreign key to match a primary key, so an order for missing customer C999 is an orphan record

A Key terms

Write the letter of the matching definition · 6 marks

- ☐ **Data dictionary**
- ☐ **Metadata**
- ☐ **Entity integrity**
- ☐ **Referential integrity**
- ☐ **Index**
- ☐ **Cascade delete**

A The rule that every table has a primary key whose values are unique and never null.

B Data that describes other data, such as a field's name, type and size.

C The rule that every foreign key value must match an existing primary key value in the related table, preventing orphan records.

D A referential integrity option in which deleting a parent record automatically deletes its related child records.

E A central store of metadata describing every table, field, data type, key, validation rule, index and relationship in a database.

F A sorted data structure of key values and pointers that lets the DBMS locate records quickly without scanning the whole table.

B Fill the gaps

Use the word bank · 5 marks

dictionary | null | index | orphan | metadata

Data about data is called _____ and is stored in the data _____. A primary key must be unique and never _____; this is entity integrity. Referential integrity prevents _____ records. An _____ speeds up searching but slows down updates.

C True or false?

Tick one box · 5 marks

	True	False
Referential integrity requires every foreign key value to match an existing primary key value.	☐	☐
A primary key may be left blank if the record is incomplete.	☐	☐
Indexes make INSERT and UPDATE operations faster.	☐	☐
Cascade delete removes child records when their parent is deleted.	☐	☐
Verification checks that data has been copied accurately from its source.	☐	☐

D Exam-style questions

16 marks

1 State three items of metadata held in a data dictionary. [3]

2 Explain what is meant by entity integrity. [2]

3 Explain how referential integrity would prevent an error in a database containing Customer and Order tables. [3]

4 Describe the difference between cascade delete and restrict delete. [2]

5 Evaluate the use of indexes in a large relational database. [4]

6 Explain why validation alone cannot guarantee that data is correct. [2]

E Quick quiz

Circle one answer · 5 marks

1 Which rule stops two customers having the same CustID? [1]

A Referential integrity
B Entity integrity
C Cascade update
D A format check

2 An Order with CustID C999 exists but there is no customer C999. This breaks... [1]

A entity integrity
B a range check
C referential integrity
D normalisation

3 Which is NOT normally stored in a data dictionary? [1]

A Field data types
B Validation rules
C The customers' addresses
D Relationships between tables

4 The main cost of adding many indexes is... [1]

A slower searches
B slower inserts and updates and more storage
C loss of referential integrity
D more redundancy in tables

5 'Qty must be between 1 and 50' is a... [1]

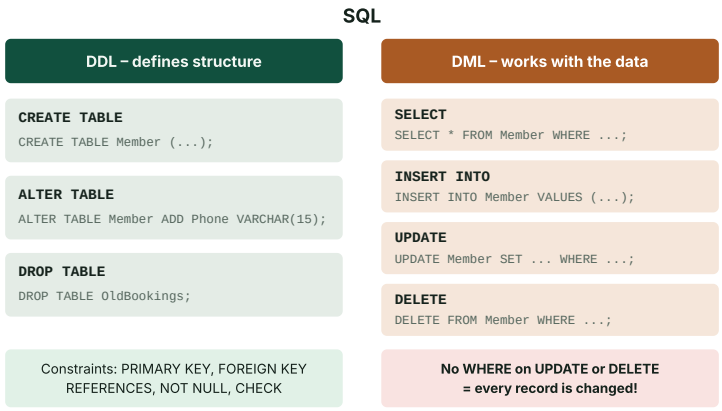
A lookup check
B presence check
C format check
D range check

SQL data definition and data manipulation

IS05 WORKSHEET

Name	Class	Date	Score

I can: Write CREATE TABLE statements with suitable data types and constraints · Write INSERT, UPDATE and DELETE statements correctly · Explain the danger of UPDATE and DELETE without a WHERE clause



Use this diagram to help you: SQL split into data definition language for structure and data manipulation language for data, with an example of each command

A Key terms

Write the letter of the matching definition · 4 marks

- ☐ **SQL**
- ☐ **Data definition language (DDL)**
- ☐ **Data manipulation language (DML)**
- ☐ **Constraint**

A A rule defined on a table, such as PRIMARY KEY, FOREIGN KEY, NOT NULL or CHECK, that the DBMS enforces on every change.

B The part of SQL used to create and change database structure, e.g. CREATE TABLE, ALTER TABLE, DROP TABLE.

C Structured Query Language: the standard declarative language for defining, querying and manipulating relational databases.

D The part of SQL used to retrieve and change data, e.g. SELECT, INSERT, UPDATE, DELETE.

B Fill the gaps

Use the word bank · 5 marks

quotes | manipulation | REFERENCES | WHERE | definition

CREATE TABLE is part of the data _____ language. INSERT, UPDATE and DELETE are data _____ commands. The _____ clause chooses which records an UPDATE affects. The FOREIGN KEY constraint uses the keyword _____ to name the parent table. Text values in SQL are written inside single _____.

C True or false?

Tick one box · 5 marks

	True	False
DELETE FROM Customer; with no WHERE clause deletes every customer record.	☐	☐

SELECT is a data definition language command.	☐	☐
VARCHAR(30) stores text of up to 30 characters.	☐	☐
A child table should be created before the parent table it refers to.	☐	☐
UPDATE changes values in existing records; it does not add new records.	☐	☐

D Exam-style questions

16 marks

1 State the difference between DDL and DML, giving one SQL command for each. [2]

2 Write an SQL statement to create a table Member with fields MemberID (6 characters, primary key), Surname (up to 30 characters, required) and JoinDate (date). [4]

3 Write an SQL statement to add a member with ID M0042, surname Corrigan, who joined on 1 September 2026. [2]

4 Write an SQL statement to increase the price of every product in the category 'Fruit' by 10%. [3]

5 A user runs DELETE FROM Booking; Explain the effect and how the database could have been protected. [3]

6 Explain how the FOREIGN KEY constraint in a CREATE TABLE statement helps maintain data integrity. [2]

E Quick quiz

Circle one answer · 5 marks

1 Which SQL command adds a new record? [1]

A UPDATE B ALTER

C INSERT INTO D CREATE

2 Which data type is most suitable for a price such as 12.49? [1]

A INT B DECIMAL(6,2)

C CHAR(5) D BOOLEAN

3 What does ALTER TABLE do? [1]

A Changes the structure of an existing table B Changes values in records

C Deletes a table D Sorts a table

4 Which clause of UPDATE says what the new values are? [1]

A WHERE B VALUES

C FROM D SET

5 Which constraint acts as a presence check? [1]

A CHECK B NOT NULL

C REFERENCES D UNIQUE

A2 1: INFORMATION SYSTEMS · SQL AND DATABASE MANAGEMENT

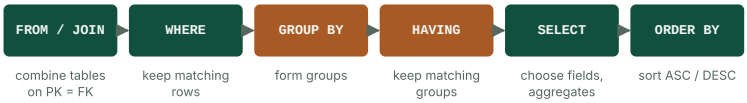
SQL queries: SELECT, aggregates, grouping and joins

IS06 WORKSHEET

Name	Class	Date	Score

I can: Write single-table SELECT queries with conditions and sorting · Use COUNT, SUM, AVG, MIN and MAX with GROUP BY and HAVING · Join two or three tables using primary key–foreign key matches

How a query is processed (logical order)



Example

```
SELECT c.CustName, SUM(ol.Qty * p.UnitPrice) AS Spent
FROM Customer c
  JOIN CustOrder o ON c.CustID = o.CustID
  JOIN OrderLine ol ON o.OrderNo = ol.OrderNo
  JOIN Product p ON ol.ProductID = p.ProductID
GROUP BY c.CustName HAVING SUM(ol.Qty * p.UnitPrice) > 100
ORDER BY Spent DESC;
```

WHERE filters rows before grouping; HAVING filters groups after aggregates are calculated

3 joins for 4 tables: n tables need n − 1 join conditions

Use this diagram to help you: The order in which a DBMS logically processes an SQL query: FROM and JOIN, WHERE, GROUP BY, HAVING, SELECT, then ORDER BY

A Key terms

Write the letter of the matching definition · 5 marks

- ☐ **Aggregate function**
- ☐ **GROUP BY**
- ☐ **HAVING**
- ☐ **Join**
- ☐ **Wildcard**

A An SQL function that calculates a single value from a set of rows, such as COUNT, SUM, AVG, MIN or MAX.

B An SQL clause that filters groups after aggregation, in the way WHERE filters individual rows.

C A symbol used with LIKE to match unknown characters: % for any sequence and _ for exactly one character.

D An SQL clause that groups rows with the same value in a field so aggregate functions are calculated for each group.

E An operation that combines rows from two or more tables by matching a foreign key with the related primary key.

B Fill the gaps

Use the word bank · 5 marks

% | aggregate | HAVING | join | WHERE

The _____ clause filters rows before grouping, while _____ filters groups. In a LIKE pattern, the _____ wildcard matches any number of characters. COUNT and AVG are examples of _____ functions. Matching a foreign key to its primary key across tables is called a _____.

C True or false?

Tick one box · 5 marks

	True	False

BETWEEN 10 AND 20 includes the values 10 and 20.	☐	☐
ORDER BY sorts in descending order unless ASC is given.	☐	☐
Joining three tables requires at least two join conditions.	☐	☐
WHERE COUNT(*) > 3 is the correct way to filter groups.	☐	☐
In LIKE patterns, _ matches exactly one character.	☐	☐

D Exam-style questions

18 marks

1 Write an SQL query to list the Surname and Forename of all pupils in YearGroup 14, in alphabetical order of Surname. [3]

2 Write an SQL query to display all products whose name begins with 'Organic' and whose price is between £1 and £3. [3]

3 Explain the difference between WHERE and HAVING. [2]

4 Write an SQL query to show each customer town and the number of customers in it, only for towns with more than five customers. [4]

5 Using tables Customer(CustID, CustName), CustOrder(OrderNo, OrderDate, CustID) and OrderLine(OrderNo, ProductID, Qty), write a query to list the names of customers who ordered product 'P07', with the order date. [5]

6 State what is returned if the join condition is omitted when querying two tables. [1]

E Quick quiz

Circle one answer · 5 marks

1 Which query finds surnames starting with Mc? [1]

A WHERE Surname = 'Mc%'
B WHERE Surname LIKE 'Mc%'
C WHERE Surname LIKE '%Mc'
D WHERE Surname IN 'Mc'

2 Which function returns the number of rows? [1]

A SUM
B AVG
C MAX
D COUNT

3 Which clause produces one result row per house? [1]

A ORDER BY House
B GROUP BY House
C HAVING House
D WHERE House

4 Customer and CustOrder are joined on... [1]

A Customer.CustID = CustOrder.CustID
B Customer.CustName = CustOrder.OrderNo
C any field with the same type
D no condition is needed

5 Which clause order is correct? [1]

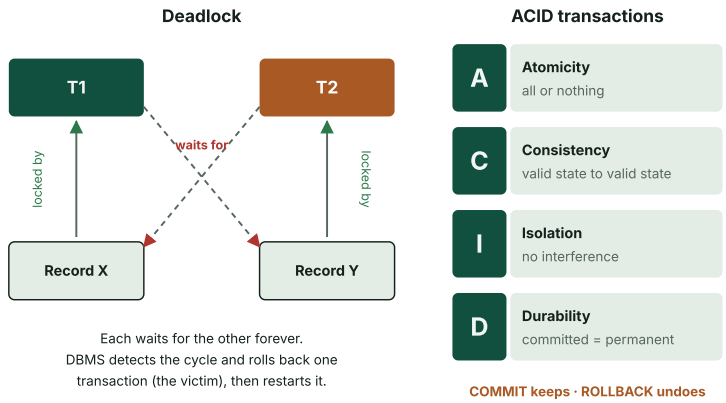
A SELECT, WHERE, FROM, ORDER BY
B SELECT, FROM, GROUP BY, WHERE
C SELECT, FROM, WHERE, GROUP BY, HAVING, ORDER BY
D FROM, SELECT, HAVING, WHERE

Multi-user databases: transactions, ACID, locking and distributed databases

IS07 WORKSHEE

Name	Class	Date	Score

I can: Explain each ACID property with an example · Describe how locking prevents the lost update problem and how deadlock can arise and be resolved · Evaluate client–server and distributed database architectures



Use this diagram to help you: Deadlock: transaction T1 holds a lock on record X and waits for Y while T2 holds Y and waits for X; the DBMS rolls one back. ACID properties listed alongside

A Key terms

Write the letter of the matching definition · 6 marks

- ☐ **Transaction**
- ☐ **ACID**
- ☐ **Record locking**
- ☐ **Deadlock**
- ☐ **Rollback**
- ☐ **Distributed database**

A A single logical database whose data is stored at several sites linked by a network but appears to users as one database.

B A sequence of database operations treated as a single logical unit that must either complete entirely or not at all.

C A concurrency control technique that prevents other users updating a record while one transaction is using it.

D A situation where two or more transactions each hold a lock the other needs, so none can proceed.

E Atomicity, consistency, isolation and durability: the properties that guarantee database transactions are processed reliably.

F Undoing all changes made by a transaction since it began, returning the database to its previous consistent state.

B Fill the gaps

Use the word bank · 5 marks

Atomicity | committed | ROLLBACK | deadlock | transaction

A _____ is treated as one unit of work. _____ means all of it happens or none of it does.

Once a transaction is _____ its changes are permanent. Two transactions waiting for each other's locks are in _____. _____ undoes a transaction's changes.

C True or false?

Tick one box · 5 marks

	True	False
Isolation means concurrent transactions do not interfere with each other.	☐	☐
Durability means a transaction can be undone at any time after it is committed.	☐	☐
Record locking can cause deadlock.	☐	☐
A distributed database has a single point of failure.	☐	☐
In a client–server database only the results of a query are sent back to the client.	☐	☐

D Exam-style questions

21 marks

1 Define the term transaction in the context of databases. [2]

2 Explain atomicity and durability, using a bank transfer as an example. [4]

3 Describe how the lost update problem can occur and how record locking prevents it. [4]

4 Explain what is meant by deadlock and how a DBMS can resolve it. [3]

5 Explain the purpose of COMMIT and ROLLBACK. [2]

6 Discuss the advantages and disadvantages of a distributed database for a chain of cinemas across Northern Ireland. [6]

E Quick quiz

Circle one answer · 5 marks

1 Which ACID property ensures a debit is never made without its matching credit? [1]

A Isolation
B Durability
C Atomicity
D Consistency

2 User A's update is overwritten by user B's update. This is the... [1]

A deadlock problem
B lost update problem
C Cartesian product
D update anomaly

3 How does a DBMS usually resolve deadlock? [1]

A Deletes both records
B Rolls back one transaction
C Commits both transactions
D Turns off locking

4 Storing copies of the same data at several sites is... [1]

A fragmentation
B normalisation
C indexing
D replication

5 Which statement makes all changes in a transaction permanent? [1]

A COMMIT
B ROLLBACK
C LOCK
D SELECT

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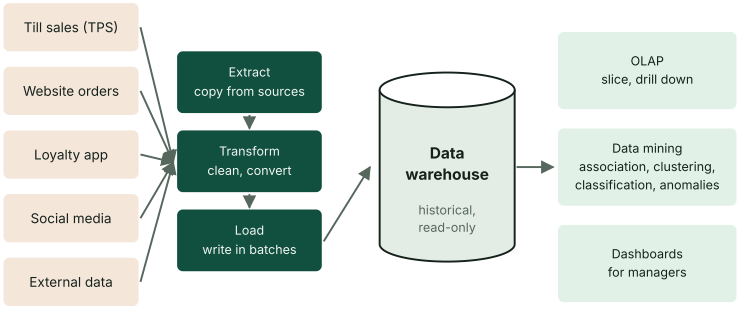
Big data, data warehousing, data mining and analytics

IS08 WORKSHEET

Name	Class	Date	Score

I can: Describe big data using volume, velocity, variety, veracity and value · Explain how a data warehouse is built and how it differs from an operational database · Evaluate the benefits of data mining for an organisation against privacy concerns

From operational data to decisions



Big data: volume · velocity · variety · veracity · value

Use this diagram to help you: Data from operational sources is extracted, transformed and loaded into a data warehouse, then analysed with OLAP and data mining to support decisions

A Key terms

Write the letter of the matching definition · 5 marks

- ☐ **Big data**
- ☐ **Data warehouse**
- ☐ **ETL**
- ☐ **Data mining**
- ☐ **OLAP**

A Data sets so large, fast-moving or varied that they cannot be processed efficiently with traditional database tools.

B Extract, transform, load: the process of copying data from source systems, cleaning and converting it, and loading it into a data warehouse.

C Online analytical processing: tools that let users analyse multidimensional data, for example by slicing and drilling down.

D A large store of integrated, historical, read-only data from many sources, organised for analysis and decision-making.

E The process of analysing large data sets to discover previously unknown patterns, relationships and trends.

B Fill the gaps

Use the word bank · 5 marks

clustering | association | transform | warehouse | volume

Big data is often described by its _____, velocity and variety. A data _____ stores historical data from many sources for analysis. Data is loaded using extract, _____, load. Discovering that customers who buy one item also buy another is called _____. Finding natural groups of customers is called _____.

C True or false?

Tick one box · 5 marks

	True	False
Velocity refers to the speed at which big data is generated and processed.	☐	☐
Data in a warehouse is normally updated many times a second by staff.	☐	☐
Data mining can discover patterns that were not previously known.	☐	☐
Anonymised data can never be linked back to an individual.	☐	☐
Data warehouses are often denormalised to speed up analytical queries.	☐	☐

D Exam-style questions

20 marks

1 Describe three characteristics of big data. [3]

2 Explain the stages of ETL used when loading a data warehouse. [3]

3 Give two differences between an operational database and a data warehouse. [2]

4 Describe two data mining techniques a supermarket could use, with an example of each. [4]

5 Discuss the benefits and privacy issues of a supermarket using loyalty-card data for data mining. [8]

E Quick quiz

Circle one answer · 5 marks

1 Which V describes the trustworthiness of data? [1]

A Volume
B Velocity
C Veracity
D Variety

2 What does the T in ETL stand for? [1]

A Transmit
B Transform
C Transfer
D Test

3 Finding unusual card transactions that may be fraud is... [1]

A anomaly detection
B association
C normalisation
D replication

4 Which is a feature of a data warehouse? [1]

A Current data only
B Fully normalised for updates
C Used for daily order entry
D Historical, integrated data for analysis

5 'What should we do?' is answered by... [1]

A descriptive analytics
B prescriptive analytics
C an index
D a transaction log

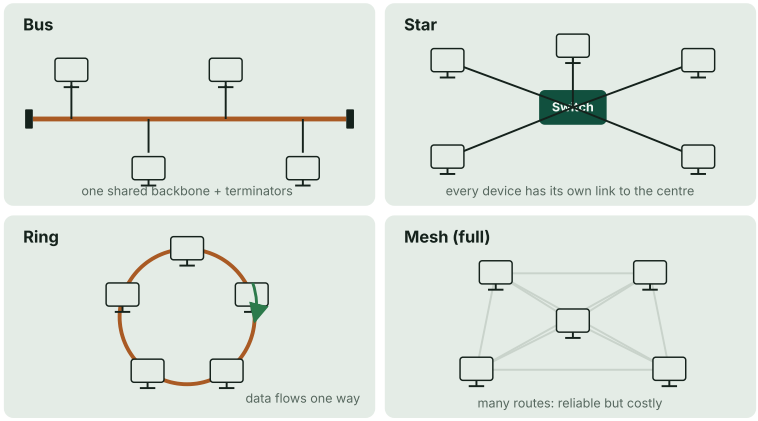
A2 1: INFORMATION SYSTEMS · NETWORKS

Network types, topologies and hardware

IS09 WORKSHEET

Name	Class	Date	Score

I can: Distinguish network types by geographical scale and ownership · Evaluate the four main topologies for a given scenario · Explain the role of each item of network hardware in moving data



Use this diagram to help you: The four network topologies: bus with a shared backbone, star with a central switch, ring as a closed loop, and mesh with multiple links

A Key terms

Write the letter of the matching definition · 6 marks

- ☐ **Local area network (LAN)**
- ☐ **Wide area network (WAN)**
- ☐ **Topology**
- ☐ **Switch**
- ☐ **Router**
- ☐ **Gateway**

A A device that connects two networks using different protocols and translates between them.

B A device that connects devices within a LAN and forwards each frame only to the port of its destination, using MAC addresses.

C A network covering a large geographical area, connecting LANs using third-party telecommunications links.

D A network covering a single site or building, with infrastructure owned and managed by the organisation.

E The physical or logical arrangement of the devices and links in a network.

F A device that connects different networks and forwards packets towards their destination using IP addresses and routing tables.

B Fill the gaps

Use the word bank · 5 marks

star | router | WAN | MAC | LAN

A network covering a single building is a _____. A network spanning several countries is a _____. In a _____ topology every device connects to a central switch. A _____ forwards packets between different networks using IP addresses. Every NIC has a unique _____ address.

C True or false?

Tick one box · 5 marks

	True	False
A switch forwards frames only to the port of the destination device.	☐	☐
In a bus topology a break in the backbone affects only one device.	☐	☐
A PAN typically connects a person's own devices over a few metres.	☐	☐
Routers use MAC addresses to forward packets across the internet.	☐	☐
A full mesh gives every node a direct link to every other node.	☐	☐

D Exam-style questions

14 marks

1 State one difference between a LAN and a WAN. [1]

2 Describe the function of a switch in a star network. [2]

3 Explain the difference between a router and a gateway. [2]

4 Give two advantages and one disadvantage of a mesh topology. [3]

5 A hotel is choosing between a bus and a star topology for its new network. Evaluate both options and recommend one. [6]

E Quick quiz

Circle one answer · 5 marks

1 Which network type covers a city? [1]

A PAN B LAN

C MAN D WAN

2 Which device lets wireless devices join a wired LAN? [1]

A Wireless access point B Gateway

C Hub D Repeater

3 In which topology does data travel around a loop from node to node? [1]

A Bus B Star

C Mesh D Ring

4 What is the main weakness of a star topology? [1]

A Collisions on the backbone B The central switch is a single point of failure

C It cannot be extended D No device has a MAC address

5 Which device translates between networks using different protocols? [1]

A Switch B NIC

C Gateway D WAP

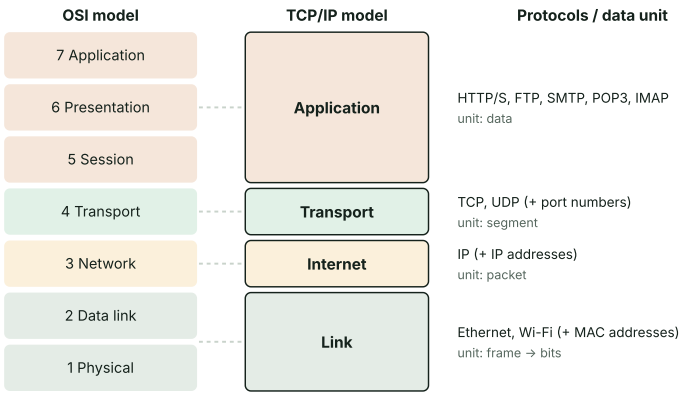
A2 1: INFORMATION SYSTEMS · NETWORKS

Layered network models and protocols

IS10 WORKSHEET

Name	Class	Date	Score

I can: Explain the benefits of a layered approach to network design · Map the OSI model to the TCP/IP model and place protocols in the correct layer · Compare TCP and UDP and choose an appropriate one for an application



Sending: down the layers, each adds a header (encapsulation). Receiving: up the layers, headers removed.

Use this diagram to help you: The seven OSI layers mapped to the four TCP/IP layers, with example protocols and the header added at each layer during encapsulation

A Key terms

Write the letter of the matching definition · 6 marks

- ☐ **Protocol**
- ☐ **OSI model**
- ☐ **TCP/IP model**
- ☐ **Encapsulation (networking)**
- ☐ **TCP**
- ☐ **UDP**

A User Datagram Protocol: a connectionless transport protocol that is fast but does not guarantee delivery or order.

B The process by which each layer adds its own header to data as it passes down the layers before transmission.

C A set of rules that governs how data is formatted, transmitted and received so devices can communicate.

D A seven-layer reference model of network communication: physical, data link, network, transport, session, presentation and application.

E The four-layer model used by the internet: link, internet, transport and application.

F Transmission Control Protocol: a connection-oriented transport protocol giving reliable, ordered delivery using acknowledgements and retransmission.

B Fill the gaps

Use the word bank · 5 marks

encapsulation | TCP | seven | UDP | protocol

A _____ is a set of rules for communication. The OSI model has _____ layers. Adding a header at each layer is called _____. _____ guarantees reliable, ordered delivery, while _____ is faster but connectionless.

C True or false?

Tick one box · 5 marks

	True	False
HTTPS uses TLS to encrypt data between the browser and the web server.	☐	☐
IMAP downloads email to one device and deletes it from the server.	☐	☐
The TCP/IP model has four layers.	☐	☐
UDP retransmits lost segments.	☐	☐
IP operates at the internet (network) layer.	☐	☐

D Exam-style questions

16 marks

1 Define the term protocol. [1]

2 Explain two advantages of using a layered model for network communication. [4]

3 Name the four layers of the TCP/IP model in order from the top. [2]

4 Describe the difference between POP3 and IMAP. [2]

5 Compare TCP and UDP and state which would be more suitable for an online multiplayer game, with a reason. [4]

6 Describe what happens to data as it passes down the layers on a sending device. [3]

E Quick quiz

Circle one answer · 5 marks

1 Which protocol sends email between mail servers? [1]

A IMAP B POP3

C SMTP D FTP

2 Which OSI layer deals with MAC addresses and frames? [1]

A Data link B Network

C Transport D Session

3 Live voice calls usually use UDP because... [1]

A it guarantees delivery B low delay matters more than the odd lost packet

C it encrypts data D it uses MAC addresses

4 Which TCP/IP layer adds port numbers? [1]

A Link B Internet

C Application D Transport

5 A key benefit of layering is that... [1]

A it removes the need for protocols B one layer can be changed without affecting the others

C data no longer needs headers D networks need no hardware

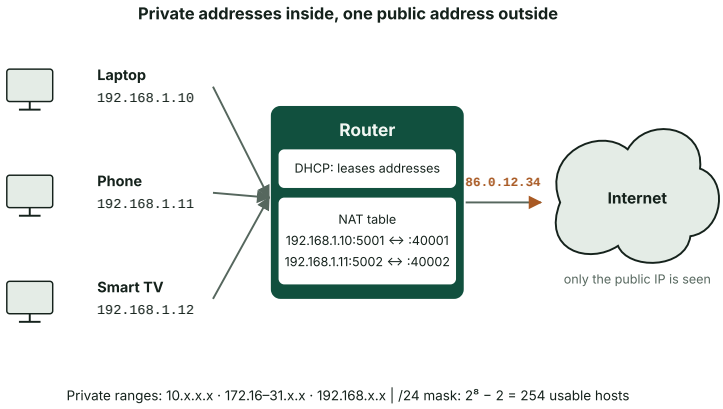
A2 1: INFORMATION SYSTEMS · NETWORKS

IP addressing, switching and routing

IS11 WORKSHEET

Name	Class	Date	Score

I can: Explain the structure of IPv4 and IPv6 addresses and why IPv6 was introduced · Explain how DHCP, NAT and subnetting are used in an organisation's network · Compare packet and circuit switching and describe how routers forward packets



Use this diagram to help you: A home network where DHCP gives devices private IP addresses and the router uses NAT to share one public IP address with the internet

A Key terms

Write the letter of the matching definition · 6 marks

- ☐ **IP address**
- ☐ **Subnet mask**
- ☐ **DHCP**
- ☐ **Network address translation (NAT)**
- ☐ **Packet switching**
- ☐ **Circuit switching**

A A method of transmission in which a dedicated path is set up and reserved between sender and receiver for the whole communication.

B A value that identifies which bits of an IP address form the network part and which form the host part.

C A logical numerical address that identifies a device on a network so data can be routed to it.

D A router technique that maps private IP addresses to a public IP address so many devices can share one public address.

E Dynamic Host Configuration Protocol: a service that automatically assigns IP addresses and network settings to devices.

F A method of transmission in which data is split into packets that travel independently across shared links and are reassembled at the destination.

B Fill the gaps

Use the word bank · 5 marks

DHCP | NAT | table | packets | 32

An IPv4 address is _____ bits long. Devices are automatically given an IP address by _____. _____ lets many devices with private addresses share one public address. On the

internet, data is split into _____ that may take different routes. Each router uses its routing _____ to choose the next hop.

C True or false?

Tick one box · 5 marks

	True	False
192.168.1.15 is a private IP address.	☐	☐
Circuit switching shares the same link among many users' packets.	☐	☐
A /24 network has 254 usable host addresses.	☐	☐
Packets always arrive in the order in which they were sent.	☐	☐
IPv6 addresses are usually written in hexadecimal.	☐	☐

D Exam-style questions

16 marks

1 State the number of bits in an IPv4 address and in an IPv6 address. [2]

2 Explain why IPv6 was introduced. [2]

3 A network uses the subnet mask 255.255.255.192 (/26). Calculate the number of usable host addresses. Show your working. [3]

4 Describe the purpose of DHCP. [2]

5 Explain how network address translation (NAT) allows several devices in a home to access the internet. [3]

6 Compare packet switching and circuit switching. [4]

E Quick quiz

Circle one answer · 5 marks

1 How many usable hosts does a /28 network have? [1]

A 16 B 14
C 28 D 30

2 Which service gives a laptop its IP address when it joins the school Wi-Fi? [1]

A NAT B DNS
C DHCP D FTP

3 What is the purpose of the time to live field in a packet? [1]

A To encrypt the packet B To stop packets circulating forever
C To record the sender's MAC address D To set the bit rate

4 Which is a feature of circuit switching? [1]

A Dedicated path for the whole call B Packets take different routes
C Links shared efficiently D Packets reassembled using sequence numbers

5 Which address is public? [1]

A 10.4.2.1 B 172.16.0.9
C 192.168.0.1 D 8.8.8.8

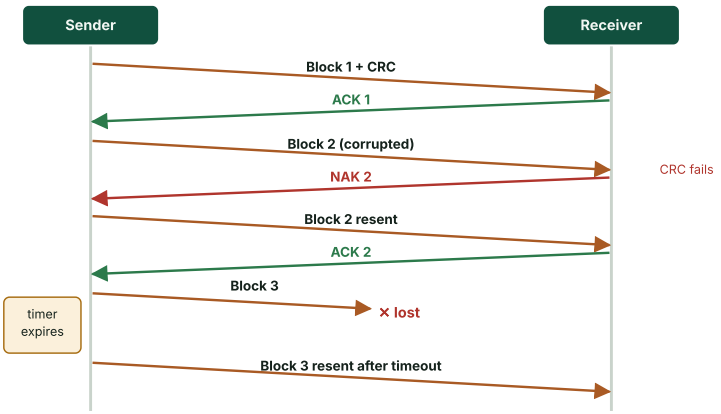
A2 1: INFORMATION SYSTEMS · NETWORKS

Data transmission and error control

IS12 WORKSHEET

Name	Class	Date	Score

I can: Compare copper, fibre-optic and wireless media for a given use · Calculate transmission times using bit rate and explain the transmission modes · Apply parity and checksum methods and explain how ARQ corrects errors



Use this diagram to help you: Automatic repeat request: block 1 is acknowledged, block 2 arrives with an error and is negatively acknowledged then resent, and block 3 is lost so a timeout triggers retransmission

A Key terms

Write the letter of the matching definition · 6 marks

- ☐ **Bandwidth**
- ☐ **Bit rate**
- ☐ **Full-duplex**
- ☐ **Asynchronous transmission**
- ☐ **Parity bit**
- ☐ **Cyclic redundancy check (CRC)**

A The capacity of a communication channel (range of frequencies), which determines the maximum rate at which data can be transmitted.

B The number of bits transmitted per second over a communication channel.

C Transmission without a shared clock, where each character is framed by start and stop bits.

D An error-detection method in which a remainder calculated by dividing the data by an agreed polynomial is sent and checked by the receiver.

E A transmission mode in which data can travel in both directions at the same time.

F An extra bit added to a group of bits so that the total number of 1s is even (even parity) or odd (odd parity), used to detect errors.

B Fill the gaps

Use the word bank · 5 marks

ARQ · checksum · parity · full-duplex · serial

Sending bits one after another along a single wire is _____ transmission. Communication in both directions at the same time is _____. An extra bit added to make the number of 1s even is a _____ bit. A _____ is a value calculated from a block of data and sent with it. _____ corrects errors by retransmitting data.

C True or false?

Tick one box · 6 marks

	True	False
Fibre-optic cable is immune to electrical interference.	☐	☐
A walkie-talkie is an example of full-duplex communication.	☐	☐
Parity checking can detect any single-bit error.	☐	☐
Asynchronous transmission uses a shared clock signal between sender and receiver.	☐	☐
Parallel transmission suffers from skew over long distances.	☐	☐
A checksum corrects errors by itself.	☐	☐

D Exam-style questions

17 marks

1 Distinguish between bandwidth and bit rate. [2]

2 Calculate how long it would take to transfer a 250 MB file over a 20 Mbps connection. Show your working. [3]

3 Add an even parity bit to 1101001 and explain one limitation of parity checking. [3]

4 Explain why serial transmission is used over long distances instead of parallel. [2]

5 Describe how automatic repeat request (ARQ) is used to correct transmission errors. [4]

6 Recommend a transmission medium to link two buildings 400 m apart on a school campus, justifying your answer. [3]

E Quick quiz

Circle one answer · 5 marks

1 How long does a 100 MB file take at 50 Mbps (1 MB = 8 Mb)? [1]

A 2 s
B 16 s
C 50 s
D 8 s

2 Which medium uses pulses of light? [1]

A Coaxial
B Twisted pair
C Fibre-optic
D Microwave

3 With odd parity, what is the parity bit for 1100110? [1]

A 1
B 0
C 2
D It cannot be calculated

4 In ARQ, what does the sender do if no acknowledgement arrives before the timeout? [1]

A Stops sending
B Sends the next block
C Asks the user
D Retransmits the block

5 Which method detects burst errors most reliably? [1]

A Parity bit
B CRC
C Echo by eye
D Simplex

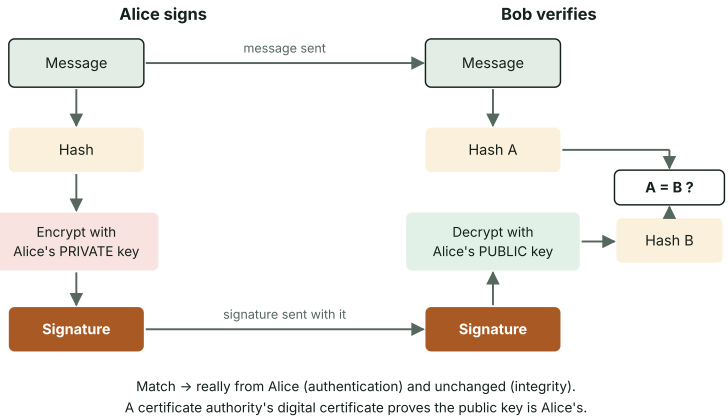
A2 1: INFORMATION SYSTEMS · NETWORK SECURITY AND CLOUD COMPUTING

Network security: threats, firewalls, encryption and authentication

IS13 WORKSHEET

Name	Class	Date	Score

I can: Describe common network threats and match each to suitable countermeasures · Explain how symmetric and asymmetric encryption, digital signatures and certificates work together · Evaluate firewalls, proxies, authentication methods and VPNs for an organisation



Use this diagram to help you: Creating a digital signature by hashing a message and encrypting the hash with the sender's private key, and verifying it by decrypting with the sender's public key and comparing hashes

A Key terms

Write the letter of the matching definition · 6 marks

- ☐ **Firewall**
- ☐ **Proxy server**
- ☐ **Symmetric encryption**
- ☐ **Asymmetric encryption**
- ☐ **Digital signature**
- ☐ **Digital certificate**

A An electronic document issued by a certificate authority that binds a public key to the identity of its owner.

B A server that acts as an intermediary between users and the internet, hiding internal addresses, filtering content and caching pages.

C Encryption in which the same secret key is used both to encrypt and to decrypt data.

D Hardware or software that monitors network traffic and allows or blocks it according to a set of security rules.

E Encryption using a mathematically related key pair: data encrypted with the public key can only be decrypted with the private key, and vice versa.

F A hash of a message encrypted with the sender's private key, used to verify the sender's identity and that the message has not been altered.

B Fill the gaps

Use the word bank · 5 marks

private | certificate | Symmetric | VPN | firewall

A _____ blocks or allows traffic according to rules. _____ encryption uses the same key to encrypt and decrypt. In asymmetric encryption, data encrypted with Bob's public key can only be

decrypted with Bob's _____ key. A digital _____ is issued by a certificate authority. A _____ creates an encrypted tunnel across the internet.

C True or false?

Tick one box · 5 marks

	True	False
A digital signature is created using the sender's private key.	☐	☐
Asymmetric encryption is faster than symmetric encryption.	☐	☐
A proxy server can cache web pages to reduce internet traffic.	☐	☐
Multi-factor authentication uses two passwords.	☐	☐
HTTPS uses both asymmetric and symmetric encryption.	☐	☐

D Exam-style questions

19 marks

1 Describe the purpose of a firewall. [2]

2 Give two functions of a proxy server. [2]

3 Explain the key distribution problem in symmetric encryption and how asymmetric encryption overcomes it. [3]

4 Describe how a digital signature is created and verified. [4]

5 State the purpose of a digital certificate. [2]

6 A firm's staff work from home. Evaluate the use of a VPN and multi-factor authentication to protect the firm's network. [6]

E Quick quiz

Circle one answer · 5 marks

1 Alice wants to send Bob a secret message. Which key does she encrypt it with? [1]

A Alice's private key
B Bob's public key
C Bob's private key
D Alice's public key

2 Who issues digital certificates? [1]

A Internet service providers
B Web browsers
C Certificate authorities
D Firewalls

3 Flooding a server with traffic from a botnet is... [1]

A phishing
B a DDoS attack
C a man-in-the-middle attack
D spyware

4 A fingerprint scan is authentication by... [1]

A something you know
B something you have
C something you are
D something you share

5 Which is NOT normally a function of a proxy server? [1]

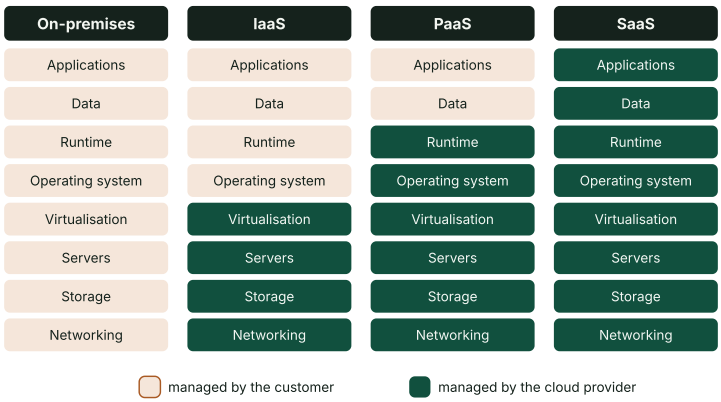
A Caching pages
B Hiding internal IP addresses
C Filtering content
D Issuing digital certificates

Cloud computing and virtualisation

IS14 WORKSHEET

Name	Class	Date	Score

I can: Distinguish IaaS, PaaS and SaaS by what the provider and customer each manage · Recommend a public, private or hybrid cloud for a scenario with justification · Explain server virtualisation and evaluate its benefits and drawbacks



Use this diagram to help you: Who manages each layer of the computing stack in on-premises, IaaS, PaaS and SaaS models: the customer or the cloud provider

A Key terms

Write the letter of the matching definition · 6 marks

- ☐ **Cloud computing**
- ☐ **Infrastructure as a service (IaaS)**
- ☐ **Platform as a service (PaaS)**
- ☐ **Software as a service (SaaS)**
- ☐ **Hybrid cloud**
- ☐ **Virtualisation**

A The on-demand delivery of computing resources such as servers, storage and software over the internet, usually paid for as used.

B A cloud model in which the provider supplies virtualised hardware such as servers, storage and networking, and the customer manages the operating system and software.

C A cloud model in which the provider supplies the hardware, operating system and development platform so customers only build and manage their own applications.

D The use of software to create virtual versions of hardware, allowing several virtual machines to run on one physical computer.

E A cloud model in which complete application software is hosted by the provider and accessed by users over the internet.

F A cloud deployment that combines a private cloud with public cloud services, allowing data and applications to move between them.

B Fill the gaps

Use the word bank · 5 marks

hypervisor | SaaS | scalability | private | service

Renting virtual servers and storage from a provider is infrastructure as a _____. Microsoft 365 is an example of _____. A cloud used by only one organisation is a _____ cloud. Software

that runs several virtual machines on one physical server is a _____. The ability to add capacity quickly as demand grows is _____.

C True or false?

Tick one box · 5 marks

	True	False
In SaaS the customer manages the operating system.	☐	☐
A hybrid cloud combines private and public clouds.	☐	☐
Virtualisation allows several operating systems to run on one physical server.	☐	☐
Using a cloud provider means an organisation no longer has any responsibilities under data protection law.	☐	☐
Cloud computing depends on a reliable internet connection.	☐	☐

D Exam-style questions

19 marks

1 Describe what is meant by software as a service (SaaS) and give an example. [2]

2 Explain the difference between IaaS and PaaS. [2]

3 Describe a hybrid cloud and give one reason an organisation might choose it. [2]

4 Explain what is meant by virtualisation and the role of a hypervisor. [3]

5 Give two benefits of server virtualisation to an organisation. [2]

6 Discuss the advantages and disadvantages for a GP surgery of storing patient records in a public cloud. [8]

E Quick quiz

Circle one answer · 5 marks

1 A developer deploys web app code without managing any servers or operating systems. This is... [1]

A IaaS
B PaaS
C SaaS
D on-premises

2 Which deployment model is shared by many organisations? [1]

A Public cloud
B Private cloud
C Hybrid cloud
D Intranet

3 Which is a drawback of cloud computing? [1]

A Scalability
B Access from anywhere
C Dependence on internet connectivity
D No upfront hardware cost

4 What does a hypervisor do? [1]

A Encrypts cloud data
B Creates and runs virtual machines
C Assigns IP addresses
D Filters web traffic

5 Data stored in another country being subject to that country's laws is an issue of... [1]

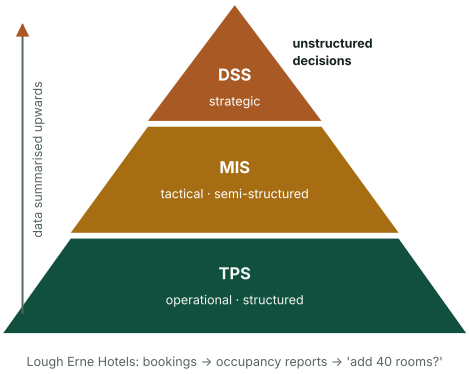
A elasticity
B virtualisation
C redundancy
D data sovereignty

Types of information system

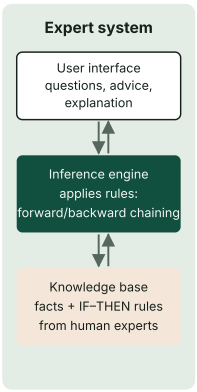
IS15 WORKSHEET

Name	Class	Date	Score

I can: Match types of information system to the levels of management they serve · Explain how a TPS, MIS and DSS work together using an example organisation · Describe the components of an expert system and evaluate its use



Lough Erne Hotels: bookings → occupancy reports → 'add 40 rooms?'



Use this diagram to help you: The information pyramid: TPS at the operational level, MIS at the tactical level and DSS at the strategic level, with data flowing upwards; and the three components of an expert system

A Key terms

Write the letter of the matching definition · 6 marks

- ☐ **Transaction processing system (TPS)**
- ☐ **Management information system (MIS)**
- ☐ **Decision support system (DSS)**
- ☐ **Expert system**
- ☐ **Knowledge base**
- ☐ **Inference engine**

A An information system that records and processes the routine day-to-day transactions of an organisation, in real time or in batches.

B An interactive information system that uses data and models to help managers make semi-structured and unstructured decisions, for example through what-if analysis.

C A computer system that emulates the decision-making of a human expert in a specific domain using a knowledge base and an inference engine.

D An information system that summarises data from transaction systems into regular, exception and on-demand reports for middle managers.

E The part of an expert system that applies the rules in the knowledge base to the facts supplied in order to reach conclusions.

F The part of an expert system that stores facts and rules gathered from human experts.

B Fill the gaps

Use the word bank · 5 marks

knowledge | inference | decision | batch | transaction

A _____ processing system records day-to-day sales and bookings. Processing all payroll records together overnight is _____ processing. A _____ support system allows

managers to carry out what-if analysis. In an expert system, the _____ engine applies rules to facts. The facts and rules are stored in the _____ base.

C True or false?

Tick one box · 5 marks

	True	False
A TPS is mainly used by directors to make strategic decisions.	☐	☐
An exception report highlights results outside normal limits.	☐	☐
Goal seeking works backwards from a desired result to find the input needed.	☐	☐
An expert system can use common sense in situations outside its rules.	☐	☐
Forward chaining starts from the known facts and works towards a conclusion.	☐	☐

D Exam-style questions

21 marks

1 State the difference between batch and real-time transaction processing, giving an example of each. [4]

2 Describe three types of report that a management information system might produce. [3]

3 Explain how a decision support system differs from a management information system. [3]

4 Describe the three main components of an expert system. [3]

5 Discuss the use of an expert system to help GPs diagnose illnesses. [8]

E Quick quiz

Circle one answer · 5 marks

1 A cinema seat booking updated immediately is an example of... [1]

A batch processing
B real-time transaction processing
C a DSS
D an expert system

2 Which system produces weekly sales reports for each branch? [1]

A MIS
B TPS
C Expert system
D Hypervisor

3 'What would profit be if we cut prices by 5%?' uses... [1]

A an exception report
B forward chaining
C the knowledge base
D what-if analysis

4 Which part of an expert system shows how a conclusion was reached? [1]

A Knowledge base
B Explanation facility
C Transaction log
D Model base

5 Who builds the knowledge base by interviewing experts? [1]

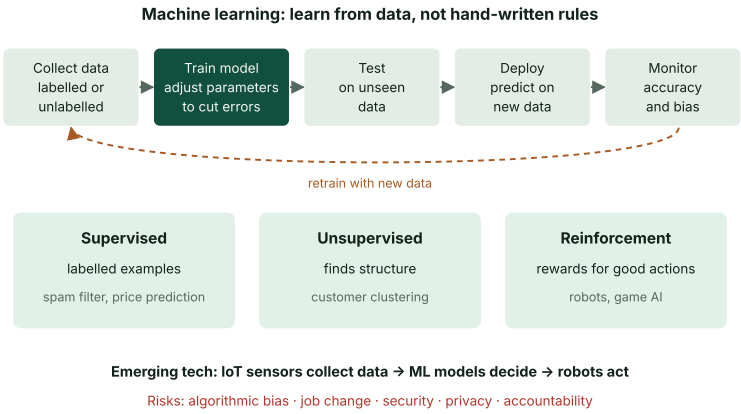
A Database administrator
B Network manager
C Knowledge engineer
D Data subject

Artificial intelligence, machine learning and emerging technologies

IS16 WORKSHEET

Name	Class	Date	Score

I can: Explain what artificial intelligence and machine learning are and how a model is trained · Describe the Internet of Things and robotics with organisational examples · Evaluate the benefits and risks of AI and emerging technologies for organisations and employees



Use this diagram to help you: The machine learning cycle of training data, training, testing on unseen data, deployment and monitoring, with the three types of learning

A Key terms

Write the letter of the matching definition · 5 marks

- ☐ **Artificial intelligence (AI)**
- ☐ **Machine learning**
- ☐ **Supervised learning**
- ☐ **Algorithmic bias**
- ☐ **Internet of Things (IoT)**

A A branch of AI in which algorithms improve their performance at a task by learning patterns from data rather than following explicitly programmed rules.

B The ability of computer systems to perform tasks that normally require human intelligence, such as recognising images, understanding language and making decisions.

C Machine learning in which a model is trained on labelled examples whose correct outputs are known.

D A network of everyday physical objects with embedded sensors, processors and connectivity that collect and exchange data.

E Systematic unfairness in the outputs of an AI system, usually caused by biased or unrepresentative training data.

B Fill the gaps

Use the word bank · 5 marks

reinforcement | Things | supervised | bias | Machine

_____ learning allows a system to learn from data rather than being explicitly programmed.

Learning from labelled examples is _____ learning. Learning through rewards and trial and error is _____ learning. Unfair results caused by unrepresentative training data are called algorithmic

_____. Everyday objects connected to the internet with sensors form the Internet of _____.

C True or false?

Tick one box · 5 marks

	True	False
All current AI systems are examples of narrow AI.	☐	☐
Unsupervised learning needs every training example to be labelled with the correct answer.	☐	☐
A model should be tested on data it was not trained on.	☐	☐
IoT devices are always secure because they are small.	☐	☐
Reinforcement learning uses rewards to improve behaviour.	☐	☐

D Exam-style questions

21 marks

1 Define machine learning. [2]

2 Distinguish between supervised and unsupervised learning, giving an example of each. [4]

3 Explain what is meant by algorithmic bias and how it can occur. [2]

4 Describe the Internet of Things and give one example of its use by an organisation. [3]

5 Evaluate the impact of introducing robots and AI in a large warehouse on the business and its employees. [10]

E Quick quiz

Circle one answer · 5 marks

1 Grouping customers into segments without predefined labels is... [1]

A supervised learning
B reinforcement learning
C unsupervised learning
D an expert system

2 Which is an IoT example? [1]

A A spreadsheet
B A smart meter sending readings to the supplier
C A printed report
D A USB stick

3 A recruitment AI rejects more women because past hiring data favoured men. This is... [1]

A overfitting
B data redundancy
C encryption
D algorithmic bias

4 What happens during training? [1]

A The model adjusts its parameters to reduce errors on training data
B The model is deleted
C Data is normalised to 3NF
D Users enter rules by hand

5 Which is a likely impact of automation on employment? [1]

A No jobs change
B Routine jobs decline and new skilled jobs appear
C All jobs disappear
D Wages always rise

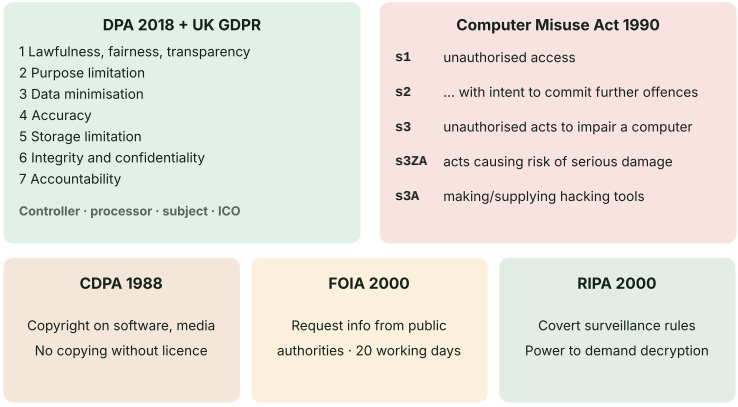
A2 1: INFORMATION SYSTEMS · LEGAL, ETHICAL AND SOCIAL ISSUES

Legislation: data protection, computer misuse, copyright, FOI and RIPA

IS17 WORKSHEET

Name	Class	Date	Score

I can: State and apply the seven data protection principles and the rights of data subjects · Identify which offence under the Computer Misuse Act applies to a scenario · Explain how copyright, freedom of information and investigatory powers legislation affect organisations and individuals



Use this diagram to help you: Summary of the five Acts: Data Protection Act 2018 and UK GDPR, Computer Misuse Act 1990, Copyright Designs and Patents Act 1988, Freedom of Information Act 2000 and RIPA 2000

A Key terms

Write the letter of the matching definition · 6 marks

- ☐ **Personal data**
- ☐ **Data controller**
- ☐ **Data processor**
- ☐ **Data subject**
- ☐ **Special category data**
- ☐ **Computer Misuse Act 1990**

A UK law creating offences of unauthorised access to computer material, access with intent to commit further offences, and unauthorised acts that impair computers.

B The living individual whom personal data is about.

C The person or organisation that decides the purposes for which, and the way in which, personal data is processed.

D A person or organisation that processes personal data on behalf of a data controller.

E Any information relating to an identified or identifiable living individual.

F Personal data needing extra protection, such as health, ethnicity, religion, political opinions, trade union membership, genetic and biometric data.

B Fill the gaps

Use the word bank · 5 marks

unauthorised · minimisation · public · ICO · controller

The organisation that decides how personal data is processed is the data _____. Collecting only the data needed follows the principle of data _____. In the UK, data protection is regulated by the _____. Section 1 of the Computer Misuse Act covers _____ access to computer material. The Freedom of Information Act applies to _____ authorities.

C True or false?

Tick one box · 6 marks

	True	False
Health data is special category data under UK GDPR.	☐	☐
The Freedom of Information Act lets anyone request information held by private companies.	☐	☐
Spreading ransomware is an offence under section 3 of the Computer Misuse Act.	☐	☐
Copyright must be registered before a program is protected.	☐	☐
Data protection law applies to data about deceased people.	☐	☐
RIPA includes a power to require encrypted data to be decrypted.	☐	☐

D Exam-style questions

18 marks

1 State four of the seven data protection principles. [4]

2 Explain the difference between a data controller and a data processor. [2]

3 A pupil uses a teacher's password to log in and change their exam marks. Identify which offences under the Computer Misuse Act 1990 have been committed and explain why. [3]

4 Describe two rights of a data subject. [4]

5 Explain the purpose of the Freedom of Information Act 2000. [2]

6 A software company discovers that a school has installed one licensed copy of its program on 50 computers. Explain which law has been broken and the possible consequences. [3]

E Quick quiz

Circle one answer · 5 marks

1 Keeping applicants' CVs for ten years with no reason breaches... [1]

A accuracy
B storage limitation
C data portability
D purpose limitation only

2 Writing and selling malware kits is an offence under... [1]

A CMA section 3A
B CDPA 1988
C FOIA 2000
D CMA section 1

3 A cloud provider storing a school's data on its behalf is a... [1]

A data subject
B data controller
C data processor
D Information Commission

4 A public authority should normally respond to an FOI request within... [1]

A 5 working days
B 40 calendar days
C 6 months
D 20 working days

5 Which Act protects software from being copied without permission? [1]

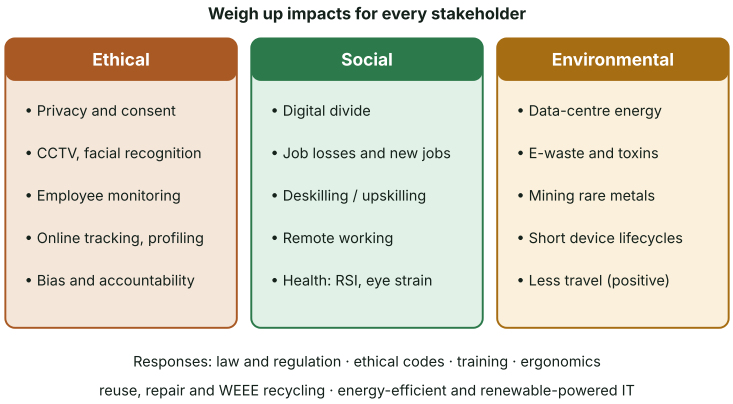
A Computer Misuse Act 1990
B Copyright, Designs and Patents Act 1988
C Freedom of Information Act 2000
D RIPA 2000

Ethical, social and environmental issues

IS18 WORKSHEET

Name	Class	Date	Score

I can: Explain privacy and surveillance issues from the viewpoints of different stakeholders · Discuss the causes and effects of the digital divide and the impact of technology on employment and health · Evaluate the environmental impact of digital technology and ways of reducing it



Use this diagram to help you: Ethical, social and environmental issues of digital technology, each with key concerns and responses

A Key terms

Write the letter of the matching definition · 5 marks

- ☐ **Digital divide**
- ☐ **E-waste**
- ☐ **Surveillance**
- ☐ **Repetitive strain injury (RSI)**
- ☐ **Sustainability**

A Discarded electrical and electronic devices, which may contain toxic and valuable materials.

B The monitoring of people's behaviour, activities or communications, for example by CCTV, tracking or interception.

C Meeting present needs without compromising the ability of future generations to meet theirs, for example by reducing energy use and e-waste.

D The gap between people who have access to digital technology and the skills to use it, and those who do not.

E Pain or damage to muscles, tendons or nerves caused by repeated movements such as typing or using a mouse.

B Fill the gaps

Use the word bank · 5 marks

e-waste | privacy | injury | deskilling | divide

The gap between those with and without access to technology is the digital _____. Discarded electronic devices are known as _____. Repeated use of a keyboard can cause repetitive strain _____. Monitoring people using CCTV and tracking raises concerns about _____. When technology does the skilled part of a job, workers may experience _____.

C True or false?

Tick one box · 5 marks

	True	False
Something that is legal is always ethical.	☐	☐
E-waste can contain toxic materials such as lead and mercury.	☐	☐
Video conferencing can reduce an organisation's carbon footprint by cutting travel.	☐	☐
The digital divide only exists between countries.	☐	☐
Regular breaks help reduce eye strain from screen use.	☐	☐

D Exam-style questions

29 marks

1 Define the term digital divide. [2]

2 Give two causes of the digital divide in Northern Ireland and suggest a way to reduce each. [4]

3 Describe two health problems that can result from prolonged computer use and how each can be prevented. [4]

4 Explain two ways an organisation can reduce the environmental impact of its IT. [4]

5 Discuss the ethical, social and environmental issues raised by an employer monitoring staff who work from home. [15]

E Quick quiz

Circle one answer · 5 marks

1 Which is a cause of the digital divide? [1]

A Fast fibre broadband in every home
B Poor rural connectivity
C Cheap devices for all
D Universal digital skills

2 Recycling electrical equipment in the UK is governed by the... [1]

A WEEE Regulations
B Computer Misuse Act
C FOI Act
D RIPA

3 Which health problem is linked to repeated mouse use? [1]

A Eye strain
B Hearing loss
C Repetitive strain injury
D Deskilling

4 A self-checkout replacing cashiers is an example of... [1]

A upskilling
B digital divide
C e-waste
D automation affecting employment

5 Which reduces the environmental impact of IT? [1]

A Replacing laptops every year
B Server virtualisation
C Leaving PCs on overnight
D Sending old devices to landfill